

EXPLAINED EXAMPLE 6 – CARNOT CYCLE

① A factory has an engine that runs a reversible Carnot cycle, detailed below. The engine runs between two temperature reservoirs at 1200 K and 300 K.

- Process 1-2: Isentropic compression
- Process 2-3: Isothermal heat addition
- Process 3-4: Isentropic expansion
- Process 4-1: Isothermal heat rejection

- ②
- a) Draw a heat engine diagram of this cycle, identifying the reservoirs and heat and work flows.
 - b) What is the thermal efficiency of this cycle?
 - c) An 800 kW heater is used to provide the heat for process 2-3. What is the net work produced by the engine?
 - d) What is the rate of change of entropy of the fluid during each of the four processes (dS/dt in kJ/K-s)?

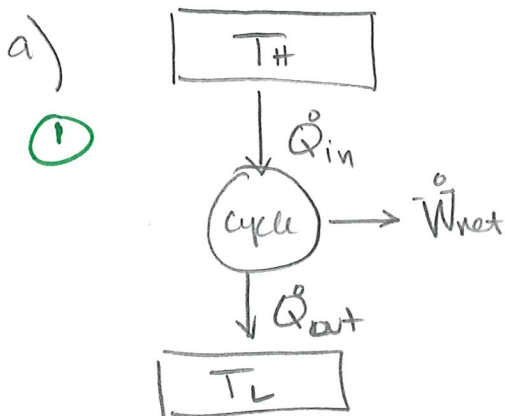
Given :

① $T_H = 1200 \text{ K}$
 $T_L = 300 \text{ K}$
 $\dot{Q}_{in} = 800 \text{ kW}$

Find :

η_{th}
 $\dot{W}_{net}$
 $\dot{S}$

Assume : ideal (reversible) cycle
 ② : quasi-steady



b) ② $\eta_{th} = \eta_{Carnot} = 1 - \frac{T_L}{T_H}$
 $= 1 - \frac{300}{1200}$

$\eta_{th} = 0.75$

c) ① $\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} \rightarrow \dot{W}_{net} = \eta_{th} \dot{Q}_{in} = (0.75)(800)$

$\dot{W}_{net} = 600 \text{ kW}$

d) Process 1-2 : isentropic compression

② $\rightarrow \boxed{\dot{S} = 0 \frac{W}{K}}$

Process 2-3 : Isothermal heat in

$$\begin{aligned}\rightarrow \dot{S} &= \int \frac{\delta \dot{Q}}{T} \Big|_{rev} \\ &= \frac{\dot{Q}_{in}}{T_H} = \frac{800}{1200} \frac{KW}{K}\end{aligned}$$

$$\boxed{\dot{S} = 0.67 \frac{KW}{K}}$$

① Process 3-4 : isentropic expansion

$$\rightarrow \boxed{\dot{S} = 0 \frac{W}{K}}$$

Process 4-1 : Isothermal heat out

$$\begin{aligned}\rightarrow \dot{S} &= \int \frac{\delta \dot{Q}}{T} \Big|_{rev} \\ &= \frac{\dot{Q}_{out}}{T_L} \rightarrow \dot{Q}_{out} = \dot{Q}_{in} - \dot{W}_{net} \\ &= 800 - 600 \\ &= \underbrace{200 KW}_{\text{heat out} < 0} \\ &= -\frac{200}{300}\end{aligned}$$

$$\boxed{\dot{S} = -0.67 \frac{KW}{K}}$$